PALEOMAGNETISM OF PRECAMBRIAN ROCKS NEAR COBALT, ONTARIO

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A paleomagnetic study was made on 32 hand samples from Precambrian rock units in the Silverfields Mine, Cobalt, Ontario. The samples from the Keewatin lavas proved unsuitable for paleomagnetic study. The Cobalt Group sediments and Nipissing diabase sill have stable primary remanences, which yield pole positions of 82.5°E, 21.5°S and 74.4°E, 11.9°N respectively after alternating-field demagnetization. The upper slate member of the Cobalt Group is in contact with the sill and its primary remanence has been altered by contact metamorphism.

INTRODUCTION

This study was undertaken to: (a) determine the suitability of the major lithologic units in the Cobalt mining camp for paleomagnetic study; and (b) study the geologic relationships between the units.

Thirty-two hand samples were collected on a vertical section through the Silverfields Mine, near Cobalt, Ontario, as shown in Fig. 1. The samples came from the steeply dipping Keewatin flows (6 samples), which form the 'basement', the flat-lying Cobalt Group sediments (13 samples), which unconformably overlie the flows, and the Nipissing diabase sill (13 samples), which intruded both flows and sediments. The samples were oriented by compass and by eye in relation to surveyed features on the mine maps, and they are estimated to be oriented within an accuracy of 3° of arc at the 95% confidence level, based on the results of this study and previous studies (e.g. Symons 1966) using the same orientation methods. Two cores (a and b) with a $1\frac{1}{4}''$ diameter and a $1\frac{3}{16}''$ height were taken from each hand sample for measurement. Samples were not collected from potentially anomalous sites such as joints, shears, blast-holes, etc.

MEASUREMENTS AND RESULTS

The density of each core was measured to within an accuracy of 0.3% by dividing the core weight in air by the difference between the core weights in air and water, as shown in Fig. 2*a*. The remanent magnetization was measured on an astatic magnetometer, described by Collinson (1967), which is sensitive to 5×10^{-8} emu/g. Measurement errors are estimated to be less than 5% in remanence intensity, and less than 2° of arc in remanence direction. Thus the total sampling and instrumental error in direction is estimated to be less than 5° of arc at the 95% confidence level. The natural remanent magnetization (n.r.m.) was measured for each core. The n.r.m. directions for all geologic groups of samples were found to be widely scattered and smeared into the present direction of the earth's magnetic field, an indication of remanence instability. Accordingly, two samples from each group were a-c. demagnetized

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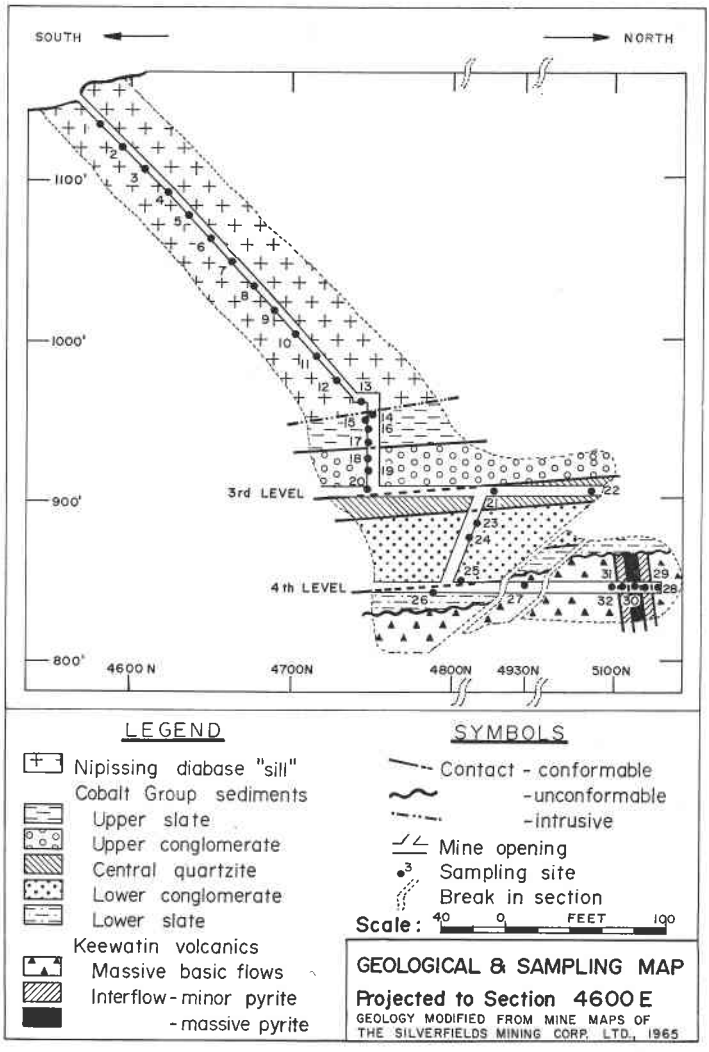


FIG. 1. Geological and sampling map of the Silverfields Mine.

in steps with peak field intensities of 70, 140, 210, 350, and 700 Oe. The alternating-field (a-c.) demagnetizing equipment has been described by Creer (1959). It demagnetizes by tumbling the specimen in an a-c. field of smoothly decreasing intensity within a field-free space created by three sets of mutually perpendicular Helmholtz coils. Calculation of the stability indices (Tarling and Symons 1967) indicated that the remaining cores of Keewatin flows should be demagnetized at a field intensity of 210 Oe, of Cobalt Group sediments at 350 Oe, and of Nipissing diabase at 280 Oe. After demagnetization, their stable remanence (*Ms*) was measured. The resulting remanence directions are shown in Fig. 3 and summarized in Table I, and the remanence intensities are shown

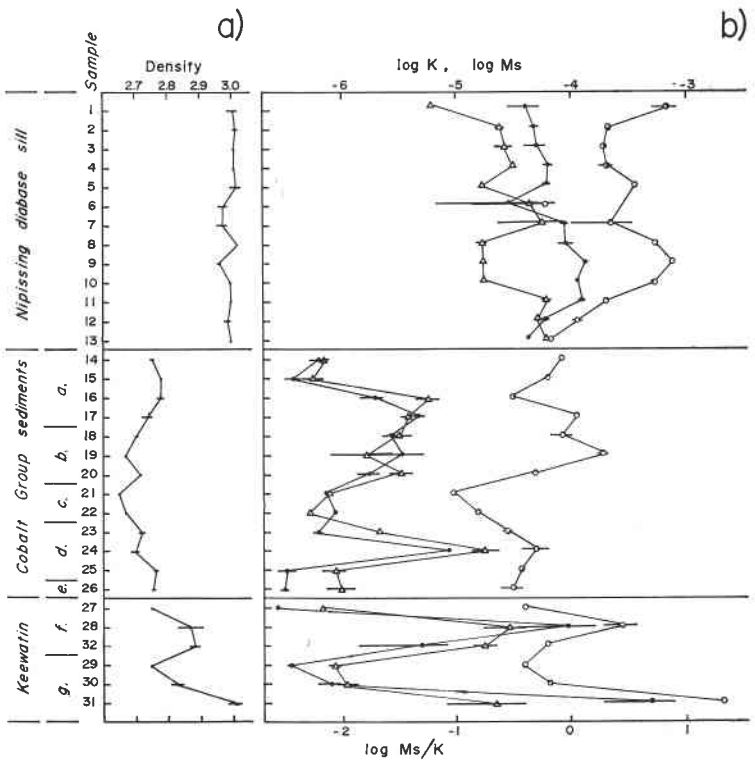


FIG. 2. Measurements on samples from: the Nipissing diabase sill; the upper slate (a), upper conglomerate (b), central quartzite (c), lower conglomerate (d), and the lower slate (e) members of the Cobalt Group sediments; and the massive flows (f) and mineralized interflows (g) of the Keewatin volcanics. The values from the two cores for each sample are shown by the ends of the horizontal line through the symbol for their mean value.
(*a*) Density in g/cm³.
(*b*) Low-field bulk susceptibility (K) (●) and remanence intensity after demagnetization (Ms) (○) in emu/g and their Koenigsberger ratio (Ms/K) (△).

in Fig. 2*b*. The bulk susceptibility K measurements were made in a low magnetic field of 10 Oe using a 'transformer-bridge type' meter, which has been described in part by Stone (1967) and in part by Collinson *et al.* (1963). The magnitude of the susceptibility is estimated to be accurate to within 5%. The bulk susceptibility measurements are shown in Fig. 2*b* with their Koenigsberger ratios (Ms/K) (Koenigsberger 1938).

STATISTICAL ANALYSIS

The statistical parameters for the remanence directions given in Table I are based on the formulae of Fisher (1953). Additional statistical tests were applied to the following groups of remanence directions: the Nipissing diabase sill, the Cobalt Group sediments, and the upper slate member. First, the null hypothesis test for randomness (Stephens 1964) shows for each of the three groups that the chance that the measured remanence directions represent a random selection from an isotropic distribution is less than 2%. Second,

TABLE I

Remanence directions and statistics

Population	Remanence						Pole position			
	D	I	N	R	k	α_{95}	Lo_E	La_N	$\delta\chi$	$\delta\psi$
Nipissing diabase	28.6	−44.6	12	11.71	37.6	7.2	74.4	11.9	9.0	5.7
Cobalt Group										
Upper slate	41.4	−30.4	4	3.34	4.6	48.2	58.9	16.2	53.7	29.9
Upper conglomerate	349.7	−74.8	3	2.81	10.5	40.1				
Central quartzite	23.4	−65.7	2	1.998	577.0	10.4				
Lower conglomerate	86.2	−65.6	2	1.93	14.0	72.7				
Lower slate	51.9	−75.0	1	—	—	—				
Overall*	35.3	−74.1	8	7.54	15.4	14.6	82.5	−21.5	26.4	23.8

*Excludes upper slate samples, which were contact metamorphosed.

D = remanence declination in degrees of arc.
I = remanence inclination in degrees of arc.
N = number of samples using average of two cores.
R = sum of the direction cosines.
k = Fisher's precision parameter.

α_{95} = radius of the cone of 95% confidence in degrees of arc.
Lo_E = Longitude east of Greenwich positive in degrees of arc.
La_N = Latitude north of equator positive in degrees of arc.
$\delta\chi$ = Semi axis (site-pole) of the oval of 95% confidence in degrees of arc.
$\delta\psi$ = Semi axis ($\perp\delta\chi$) of the oval of 95% confidence in degrees of arc.

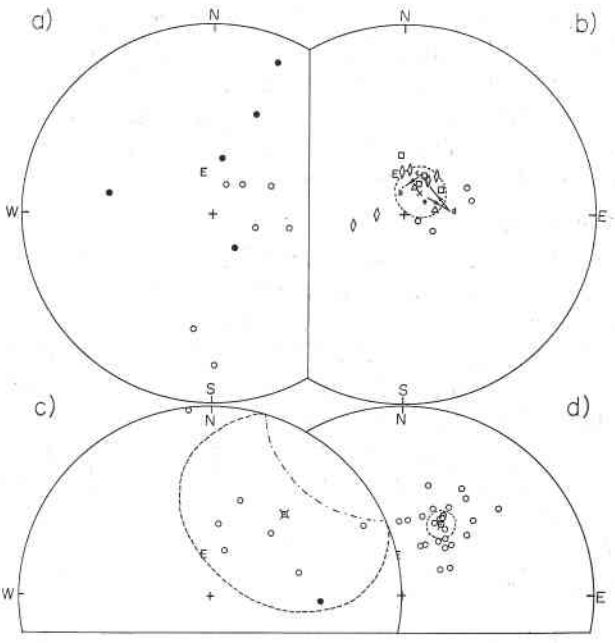


FIG. 3. Remanence directions before tectonic correction for both cores of each sample from the:

(*a*) Keewatin volcanic flows and mineralized interflows.

(*b*) Cobalt Group sediments—upper conglomerate (◇ (b)), central quartzite (□(c)), lower conglomerate (○ (d)), and lower slate (△ (e)). The lettered dots joined by the solid line represent the mean directions for the members and indicate the apparent change in orientation of the earth's magnetic field through apparent polar wander during deposition.

(*c*) Upper slate member.

(*d*) Nipissing diabase sill.

The remanence directions are plotted on Wulff stereographic projections with their declinations reckoned clockwise from north and their positive (negative) or downwardly (upwardly) directed inclinations shown by solid (open) symbols. The mean remanence direction is indicated by a cross through the group symbol or by a cross alone and the corresponding cone of 95% confidence shown by a dash-dotted (dashed) line. The E indicates the present positive direction of the earth's magnetic field.

Snedecor's F-ratio test for within-group dispersion of vectors (Larochelle 1967) shows for each of the three groups that there is less than a 5% chance that its directions were drawn from a population dispersed identically to either of the remaining two groups. Third, the angular variance estimates test for between-group dispersion (Larochelle 1967) shows that there is less than 0.01% chance that the Nipissing sill and Cobalt sediment directions were drawn from the same population, a 0.9% chance for the Cobalt sediment and upper slate member, and a 20% chance for the upper slate member and Nipissing sill. Clearly, the remanence for each group is real and distinctive, and reflects its differing genetic history. Further, the remanence of the upper slate member is much more akin to that of the Nipissing sill than to that of the Cobalt sediments.

DISCUSSION OF RESULTS

Keewatin Flows

The Keewatin flows form the 'basement' at the Silverfields Mine (Fig. 1). They are steeply folded and regionally metamorphosed greenstones with

pyritiferous, tuffaceous, interflow layers. The remanence directions (Fig. 3*a*) and other magnetic properties (Fig. 2*b*) of the samples from the Keewatin flows do not show either within-sample or between-sample consistency. The a-c. demagnetizing tests also indicate metastable remanence as both samples yielded very low stability-index values. Hence, the Keewatin flows are probably not suitable for additional paleomagnetic work.

Cobalt Group Sediments

The Cobalt Group sediments are nearly flat-lying at the Silverfields Mine, with dips of less than a few degrees in various directions. They unconformably overlie the Keewatin flows. The maximum age of the sediments is indicated by one K-Ar radiometric age date of 2605 m.y. (Lowdon *et al.* 1963) and one Rb-Sr date of 2670 ± 80 m.y. (Van Schmus *et al.* 1963) on older granitic rocks that cut the basement rocks. The upper slate member of the Cobalt Group has been contact metamorphosed. The effects of this metamorphism on the remanence are discussed below. The samples from the remaining members present (Fig. 1) are impure quartzites, grits, graywackes or granular conglomerates, except for one sample (number 24), which is a pebble conglomerate. Wilson (1913), Tagseth (1959), Thomson (1961), and Barker (1966) reported that detrital magnetite or titanomagnetite is the major ferromagnetic mineral, with minor hematite, and that the clastic grains were derived from a wide variety of volcanic and clastic rocks.

The Cobalt sediments probably have a stable primary remanence because: (1) the stability-index values from a-c. demagnetization tests indicate moderate stability; (2) the high degree of within-sample and between-sample consistency of remanence directions (Table I; Fig. 3*b*); (3) the widely divergent (150° of arc apart) and isolated directions given by the two cores from the pebble conglomerate sample are thought to reflect randomly oriented pebbles, which is the 'conglomerate test' of Graham (1949); and (4) the mean remanence direction for each sedimentary member traces a regular path on the stereonet (Fig. 3*b*), which suggests apparent polar wander of the earth's magnetic field during deposition. The primary remanence is probably a detrital remanent magnetization (d.r.m.) or chemical remanent magnetization (c.r.m.) that was acquired during sedimentation or diagenesis. The mean stable remanence direction of 35.3°, −74.1° (α_{95} = 14.6°) is thought to reflect the earth's field at the time of deposition or diagenesis, and to provide a good estimate using the dipole model of the pole position (82.5° E, 21.5° S) at that time. This pole position is similar, though reversed, to positions reported by Symons (1966) for nearly contemporaneous Animikie iron-formations of the Lake Superior region.

Nipissing Diabase Sill

The Nipissing quartz diabase intruded the Cobalt Group sediments and Keewatin flows as a sill several hundreds of feet thick. The time of intrusion is indicated by one K–Ar radiometric age date of 2095 m.y. (Lowdon *et al.* 1963) and six Rb-Sr dates on feldspar and the whole rock of 2170 ± 80 m.y. (Van Schmus *et al.* 1963). These ages differ radically from the value of about 1200

m.y. postulated by Strangway (1964) on the basis of the rate of decay of stable remanence intensity. The lack of evidence of significant structural deformation, the underlying horizontal sediments, and the fresh appearance of the diabase favor retention of an unaltered primary remanence. The remanence of the diabase samples is associated with titanomagnetite, which is the only significant ferromagnetic mineral (Satterly 1928).

One sample (number 1) of Nipissing diabase from near the surface had an extremely intense n.r.m., anomalous remanence directions and an abnormally high Koenigsberger ratio. It may have been lightning struck and hence was discarded. The remanence of the remaining samples is probably a primary thermal remanent magnetization (t.r.m.) acquired during cooling because it yields a 'stable' stability-index value upon a-c. demagnetization and a high degree of within-sample and between-sample consistency in its magnetic properties (Table I; Figs. 2 and 3*d*). Strangway (1964) reported a remanence direction of 180°, +65° for one sill sample. This is nearly an 180° reversal of the direction found here, which is additional evidence for remanence stability. This evidence for remanence stability suggests that the calculated pole position (74.4°E, 11.9°N) reliably reflects the earth's field at the time the diabase was intruded.

Upper Slate Contact Zone

At the Silverfields Mine, the upper slate member of the Cobalt Group sediments immediately underlies the Nipissing diabase sill. The samples from the upper slate have fairly stable remanences as shown by the 'poorly stable' stability-index value yielded by a-c. demagnetization, and by the within-sample consistency of their remanence intensities and susceptibility magnitudes (Fig. 2*b*). However, the remanence directions for this group show a moderately high within-site and high between-site scatter, as shown in Fig. 3*c* and Table I. Also, their mean remanence direction aligns most closely with that of the Nipissing sill rather than that of the primary Cobalt Group sediments. These magnetic properties suggest that the remanence of the upper slate member is composed of two or more components. Probably, the primary d.r.m. or c.r.m. of the upper slates has been mostly obliterated by contact metamorphism during intrusion of the Nipissing sill, which impressed a secondary remanence on the slates. This secondary remanence is probably a t.r.m. associated with heating to a temperature of about 500 ± 75 °C, which is the probable Curie Point range for the primary ferromagnetic minerals, or possibly a c.r.m. associated with hydrothermal chemical reactions. Detailed mineralogic studies of the ferromagnetic minerals should distinguish between the t.r.m. and c.r.m. alternatives. It does suggest that a 500 ± 75 °C contact metamorphic isotherm could be drawn at or near the base of the upper slate or 20 to 25 ft below the Nipissing sill. Barker (1966) placed a 300 °C isotherm at about the same distance on petrologic evidence.

CONCLUSIONS

1. The paleomagnetic results are consistent with the sequence of geologic events proposed by Thomson (1961) for this area. First the Keewatin flows were deposited and then folded. Next, the Cobalt sediments were deposited

unconformably over the flows. Finally, the Nipissing diabase sill intruded the sediments and flows and produced a zone of contact metamorphism.

2. The Keewatin flows and mineralized interflows do not appear to be suitable for paleomagnetic study.

3. The Cobalt Group sediments and Nipissing diabase sill have stable primary d.r.m. or c.r.m. and t.r.m. respectively. They yield reliable pole positions of 82.5°E, 21.5°S and 74.4°E, 11.9°N for the earth's magnetic field in the early Aphebian (Stockwell 1964) and mid-Aphebian (about 2095 m.y.) era respectively. These pole positions are subject to undetermined tectonic corrections of less than the radius of their ovals of 95% confidence. The angular divergence of 34° of arc between these pole positions suggests a time interval in the order of 10^7 to 10^8 years between deposition of the sediments and intrusion of the sill based on known rates of apparent polar wander of about 20° of arc per 100 m.y. (Irving 1964). This is consistent with the radiometric age determinations.

4. The upper slate member of the Cobalt Group has a stable secondary t.r.m. or c.r.m. acquired through contact metamorphism during intrusion of the Nipissing diabase sill. Thus the 500 ± 75 °C isotherm of contact metamorphism is at or near the base of the upper slate or 20 to 25 ft below the sill.

Further paleomagnetic studies are planned to locate the pole positions and isotherms more exactly, and to study the structure of the Cobalt area.

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